

TotalNo.ofQuestions:10]

SEATNo.:

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[6351]-114

FirstYearofEngineering

BASICELECTRONICSENGINEERING

(2024Pattern)(Semester-I)(ESC-101-ETC)

Time:2½Hours]

[Max.Marks:70

InstructionstotheCandidates:

- 1) AttemptQ.1orQ.2,Q.3orQ.4,Q.5orQ.6,Q.7orQ.8,Q.9orQ.10.
- 2) Neatdiagramsmustbedrawnwherevernecessary.
- 3) Figurestothe right indicate full marks.
- 4) Assumesuitable data, if necessary.
- 5) Use of electronic pocket calculator is allowed.

- Q1)a) Explain the Impact of Electronics on industry and society? [5]
- b) Draw circuit diagram and explain operation of Bridge Rectifier with the help of neat waveforms? [5]
- c) Compare Active and Passive Components. [4]

OR

- Q2)a) Draw circuit diagram and explain operation of Half-Wave Rectifier with the help of neat waveforms? [5]
- b) Draw and explain V-I characteristics of P-N Junction Diode. [5]
- c) Compare LED and Photodiode. [4]

- Q3)a) What is N-Well method of VLSI CMOS manufacturing. [5]
- b) Draw the construction diagram and explain the operation of Enhancement type N-channel MOSFET in detail? [5]
- c) Determine the d.c. current gain β (Beta) and the emitter current I_E for a transistor where $I_B = 100\mu A$ and $I_C = 5.65mA$. [4]

OR

P.T.O.

- Q4)a)** Draw and explain MOSFET as Switch. [5]
- b) Draw and explain the Output Characteristics of BJT in common emitter configuration and explain in detail? [5]
- c) Determine the dc current gain β (Beta) and the emitter current I_E for a transistor where $I_B = 50\mu A$ and $I_C = 3.65mA$. [4]

- Q5)a)** Classify the different types of Logic Gates? Draw and explain any two Logic gates with its truth table & logic equation. [5]
- b) Draw the block diagram of Microprocessor and explain the functions of each block in detail. [5]
- c) Convert: [4]
- i) $(1100111)_2$ into $(--)_10$
- ii) $(75.371)_{10}$ into $(--)_2$

OR

- Q6)a)** State and Prove De-Morgan's Theorem. [5]
- b) Draw & Explain Full Adder Circuit with truth table. And logical expression. [5]
- c) Compare Microprocessor and Microcontroller in detail. [4]

- Q7)a)** Draw the block diagram of Digital Multimeter, Explain the function of each block. [5]
- b) Draw the Functional block diagram of operational amplifier and explain in detail. [5]
- c) Write any two Ideal and Practical characteristics of Op-Amp IC 741 [4]

OR

- Q8)a)** Draw the block diagram of Function Generator. Explain the function of each block. [5]
- b) For inverting amplifier using op-amp. If $R_1 = 1k\Omega$ and $R_f = 10k\Omega$ with $V_{CC} = \pm 12V$ applied calculate output voltage if, i) $V_{in} = 20mV$
- ii) $V_{in} = 2.2V$ Justify answer [5]
- c) Draw and explain the block diagram of Regulated DC power supply. [4]

- Q9)**a) What is the Selection criteria used while selecting the sensors for particular application? [5]
- b) Draw and explain Electromagnetic Frequency Spectrum with their applications. [5]
- c) Comparison of Active Sensors and Passive Sensors. [4]

OR

- Q10)**a) Describe the Block diagram of IoT based Data Acquisition. [5]
- b) Describe the Block diagram of GSM. [5]
- c) Describe the Basic block diagram of Communication System. [4]



